

Day 23 Quiz

Geometry Summer Credit | Modeling With Geometry

Name:	Date:	Score:
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Directions

Choose the best answer. Show work for modeling decisions.

1.	The first step in a geometry modeling problem is usually to: A calculate immediately B understand the situation and assumptions C ignore units D choose the largest formula
2.	To decide how much wrapping paper covers a box, the best measurement is: A perimeter B area or surface area C volume D density
3.	To decide how much liquid a tank holds, the best measurement is: A volume B surface area C angle measure D perimeter
4.	A scale factor of 3 changes volume by a factor of: A 3 B 6 C 9 D 27
5.	A constraint is: A a required limit or condition B always the final answer C a unit conversion only D never part of a model
6.	A rectangular package is 12 in by 10 in by 6 in. Find its volume and decide if it meets a minimum capacity of 700 cubic inches.
7.	A wall is 28 ft by 9 ft. Paint covers 300 square feet per gallon. How many gallons are needed for one coat?

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Continued

8.	A cylindrical container has radius 4 cm and height 8 cm. Find its volume in terms of π .
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9.	Name one assumption you might make in item 7 and explain how it affects the model.
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10.	Error analysis: A student finds the area of the base of a box to decide whether it holds enough cereal. Explain the error.
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